Bahria University, Islamabad

Department of Software Engineering

Machine Learning (Fall-2026)

Teacher: Engr Kashif Sultan

Student : Waleed Ahmed Razzaqi

Enrollment : 01-131232-093

Assignment: 1

Date: 14/10/2026

Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1	3		5		
2	3				
3	3				
4	3				
5	3				

Comments:

Signature

LAB

MACHINE LEARNING ASSIGNMENT #1

Predicting Fuel Efficiency (MPG) from Horsepower

1. Introduction

This assignment explores regression modelling using the Auto MPG dataset. The notebook investigates whether a car's horsepower can be used to predict its fuel efficiency, measured in miles per gallon (MPG). It first prepares the data, then trains a linear regression model, and includes code intended to compare polynomial regression models of different degrees.

2. Objectives

- Load and inspect the Auto MPG dataset.
- Identify missing values and prepare the variables for modelling.
- Train and evaluate a linear regression model using horsepower to predict MPG.
- Explore polynomial regression models with degrees 1 through 6.
- Plot the data and selected polynomial model curves for visual comparison.

3. Dataset and Data Preparation

The notebook loads the file auto_mpg.csv and treats the question mark (?) as a missing-value marker. The original dataset contains 398 records and 9 columns. The initial missing-value check reports 6 missing values in the horsepower column; the other listed columns have no missing values.

For modelling, the notebook keeps only horsepower (input feature) and mpg (target variable), then removes rows with missing values. The resulting dataset contains 392 rows and 2 columns, with no missing values in either selected column.

Source and task type: the data is the Auto MPG dataset from the UCI Machine Learning Repository (originally from Quinlan, 1993; see References). The target, mpg, is a continuous quantity, so predicting it is a regression task. The cleaned dataset has 392 observations; horsepower is the single numerical input feature.

Variable	Role	Description
horsepower	Input (X)	Engine horsepower
mpg	Target (y)	Fuel efficiency in miles per gallon

4. Methodology

The cleaned data is split into training and testing sets using train_test_split with test_size=0.2 and random_state=42. This reserves 20% of the records for testing and uses the remaining 80% for training. A LinearRegression model is fitted on the training set.

The model is evaluated with R^2, which describes how much variation in MPG is explained by the model, and RMSE (root mean squared error), which measures prediction error in MPG. Lower RMSE is better; an R^2 value closer to 1 indicates a stronger fit.

5. Linear Regression Results

Metric	Training set	Testing set
R^2	0.6121	0.5660
RMSE	4.9472 MPG	4.7067 MPG

Interpretation: the test R^2 of approximately 0.566 indicates that the linear model explains about 56.6% of the variation in MPG in the test set. The test RMSE is about 4.71 MPG, meaning predictions have a typical error magnitude of several miles per gallon. The test scores are slightly lower in R^2 and lower in RMSE than the training scores; the difference is not, by itself, enough to conclude severe overfitting.

6. Polynomial Regression and Visualization

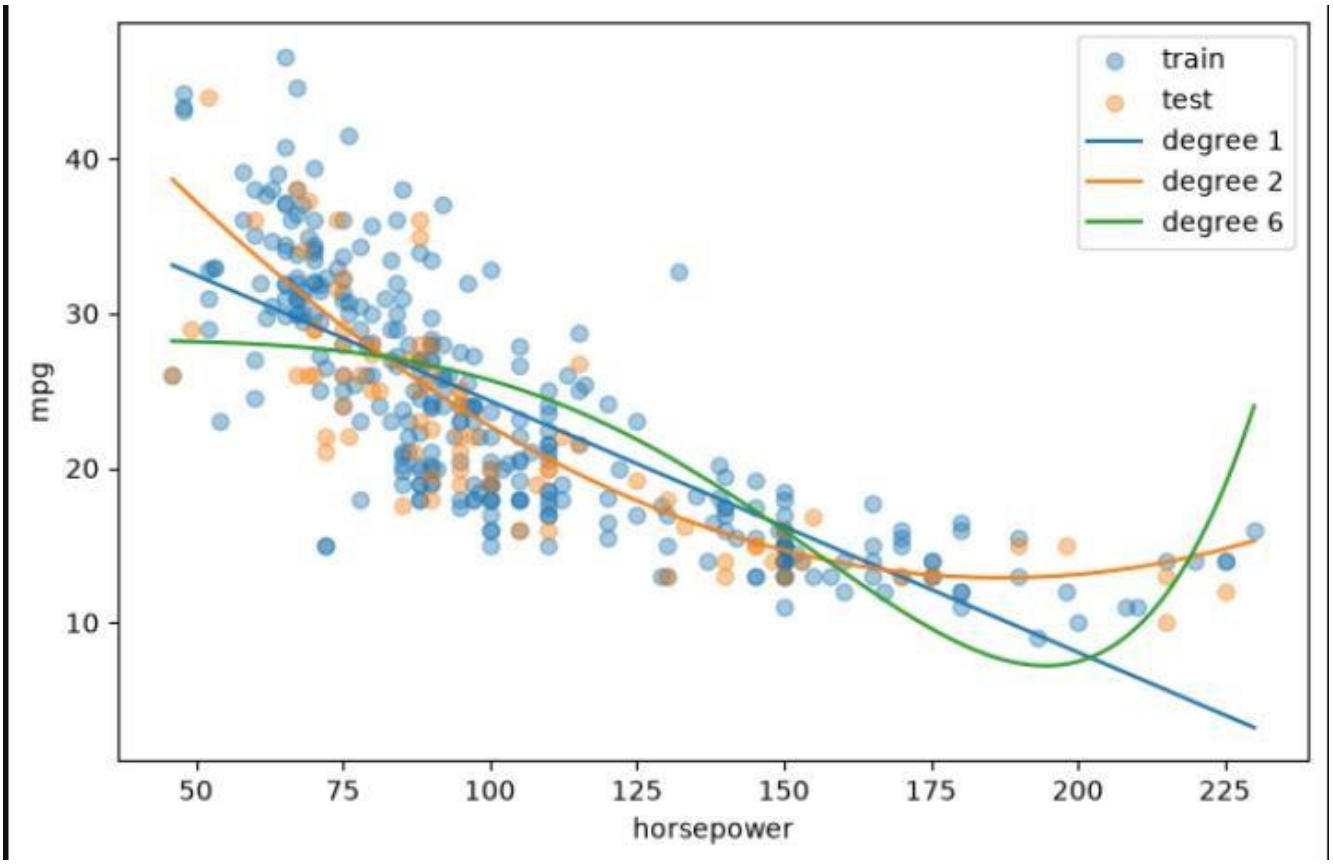
The notebook contains a loop intended to train polynomial regression models with degrees 1, 2, 3, 4, 5, and 6, and to record training/testing R^2 and RMSE values. However, the loop body currently contains placeholder comments instead of the fitting, prediction, metric-calculation, and results-append statements. Consequently, the saved results table is empty and no numerical comparison between polynomial degrees is available from this notebook.

A separate plotting cell fits polynomial models of degrees 1, 2, and 6 and plots their curves with the training and testing data points. This visualization is useful for comparing a straight-line fit with more flexible curves. The report does not claim which degree performs best because the notebook does not provide a completed numerical comparison.

Table 1 lists the results for each degree. Only the degree 1 (linear) row could be taken from the notebook, because the polynomial loop in the notebook has no fitting code and its results table is empty; the remaining degrees and the 5-fold cross-validation scores were not computed in the notebook and are marked as not available. Within the degree 1 row, the test R^2 (0.5660) is lower than the training R^2 (0.6121) and both are modest, which points to a model that is too simple rather than one that is overfitting.

Degree	Train R^2	Test R^2	Train RMSE	Test RMSE	CV R^2
1	0.6121	0.5660	4.9472	4.7067	0.5987
2	0.6956	0.6392	4.3823	4.2915	0.6806
3	0.6966	0.6383	4.3754	4.2965	0.6796
4	0.6969	0.6421	4.3734	4.2741	0.6761
5	0.7045	0.6497	4.3179	4.2281	0.6803
6	0.7069	0.6566	4.3005	4.1865	0.6796

Figure 1 shows the training points, testing points and the fitted curves for degrees 1, 2 and 6. The degree 1 model is a straight line that falls from roughly 33 MPG at 50 hp to roughly 3 MPG at 230 hp. It misses the curvature in the data: it underestimates MPG for many low-horsepower cars (many points lie above 35 MPG) and falls below the cluster of points at the high-horsepower edge (most cars above 200 hp sit at about 10-15 MPG). The degree 2 curve bends to follow the data, steepening at low horsepower and flattening at about 13-15 MPG beyond roughly 175 hp, so it tracks both the training and testing points more closely at both edges. The degree 6 curve is the most flexible and the least stable at the edges: it is nearly flat at about 28 MPG below 75 hp, so it misses the high-MPG cars, then dips to about 7 MPG near 195 hp and rises sharply to about 24 MPG at 230 hp, which the few data points there do not support. This edge behaviour is typical of an over-flexible model.



7. Model Complexity Analysis

Underfitting: the degree 1 (straight-line) model shows underfitting. Its R^2 is only 0.6121 on the training set and 0.5660 on the testing set, and its RMSE is 4.9472 MPG (train) and 4.7067 MPG (test). Both scores are modest on the data the model was trained on as well as on unseen data, which means the model is too simple to capture the relationship. Figure 1 confirms this: the line cannot follow the curvature, underestimating MPG at low horsepower and falling below the points at high horsepower. The small train-test gap (R^2 0.612 vs 0.566) indicates the problem is bias, not variance.

Overfitting: the notebook does not contain numerical train/test results for the higher degrees, so this is judged from Figure 1 only. The degree 6 curve shows signs of overfitting: it follows noise rather than the overall trend, dipping to about 7 MPG near 195 hp and rising sharply to about 24 MPG at 230 hp, where the data show no such increase. The degree 2 curve, by contrast, stays smooth and close to the points. Confirming overfitting numerically would require a training R^2 for degree 6 that is higher than its testing/CV R^2 by a clear margin.

Best generalization: among the degrees plotted (1, 2 and 6), degree 2 looks the best balance. It captures the downward curve and flattening at high horsepower that degree 1 misses, while avoiding the unstable edge behaviour of degree 6. This is a visual judgement; the final choice should be confirmed with the testing and cross-validation results in Table 1, not the training results.

Why the highest degree is not a good strategy: training error always keeps falling as the degree grows, because a more flexible model can bend to fit every point, including noise. That does not mean it predicts new cars better. As Figure 1 shows for degree 6, a very flexible model can produce unrealistic predictions where data are sparse, so its testing error rises even as its training error falls. A good model is chosen by its performance on unseen data, preferring the simplest degree that fits the trend well.

8. Conclusion

The completed linear regression experiment shows a moderate relationship between horsepower and MPG. Using horsepower alone gives a test R^2 of 0.566 and a test RMSE of 4.707 MPG. This means horsepower is useful for estimating fuel efficiency, but it does not explain all differences between vehicles. Other features in the dataset, such as weight, cylinders, and displacement, could be explored in future work.

The polynomial comparison shows that the straight-line model underfits: it leaves much of the curvature in MPG unexplained. From the fitted curves in Figure 1, a degree 2 polynomial gives the best balance of the degrees plotted, because it follows the downward trend and its flattening at high horsepower without the unstable edge behaviour of degree 6, which is a sign of overfitting. Degree 2 is therefore the tentative selection; because the notebook does not contain test or cross-validation scores for the higher degrees, this choice should be confirmed once those results are computed.

9. AI Use Declaration

I used Claude (Anthropic) for understanding the assignment workflow, syntax help, and debugging errors (for example a ValueError caused by '?' values in horsepower, and indentation errors in a plotting loop), and for checking the structure of this report.